APPENDIX E. Abbreviated Historical Site Assessment (HSA)

ABBREVIATED HISTORICAL SITE ASSESSMENT (HSA)

This section provides an abbreviated NMRC Historical Site Assessment (June 1999). This HSA follows closely the guidance provided in MARSSIM, Chapter 3 (reference 10.1). In the MARSSIM, the Radiation Survey and Site Investigation (RSSI) Process uses a graded approach that starts with the Historical Site Assessment (HSA) and is later followed by other surveys that lead to the final status survey. In some cases, where sealed sources or small amounts of radionuclides are described by the HSA, the site may qualify for a simplified decommissioning procedure.

E-1. EXECUTIVE SUMMARY

The purpose of this Historical Site Assessment (HSA) is to provide documentation useful in determining the level of effort necessary to fulfill the radiological decommissioning requirements of the U. S. Nuclear Regulatory Commission (NRC).

Base Realignment and Closure (BRAC) legislation passed and signed into law in 1995 mandated that the Naval Medical Research Center (NMRC) (formerly Naval Medical Research Institute) relocate some of its programs and research efforts and cease operations at the National Naval Medical Center (NNMC) campus in Bethesda, Maryland by the end of October 1999. NMRC, the Navy's largest medical research facility, opened its doors in Bethesda in October 1942 to conduct research, development, tests, and evaluations to enhance the health, safety, and readiness of Navy and Marine Corps personnel in the effective performance of peacetime and contingency missions. As a Tenant Command on the NNMC campus, NMRC has specific environmental obligations to fulfill before vacating the NNMC buildings it currently occupies. One of those obligations is the radiological decommissioning of those NNMC buildings and the leased Rockville Annex facility that were used to conduct research and development missions.

NMRC will be among the first Navy commands to undergo a facility decommissioning using the Nuclear Regulatory Commission's (NRC) dose-based criteria for radiological release for unrestricted future use and the guidance provided in the MARSSIM (Multi-Agency Radiation Survey and Site Investigation

Manual, NUREE-575, December 1997). NMRC's decommissioning efforts will include the preparation of this Historical Site Assessment, the development and implementation of a Radiological Decommissioning Plan, clearance of equipment and materials, and the transfer of operations involving licensed-materials use from a Navy Radioactive Material Permit to an Army installation's NRC license.

This document should not be used as a complete historical accounting of the research efforts and activities conducted at NMRC. This document focuses on those aspects closely or directly related to NMRC's acquisition, use, storage and disposal of licensed (radioactive) materials during the period 1942 through 1999. All available and pertinent NMRC files and documentation have been examined with the purpose of preparing this Historical Site Assessment. Data and findings were summarized and recommendations are provided regarding decommissioning the NMRC sites at Bethesda and at the Rockville Annex. The potential radioactive contaminants include Hydrogen-3 (tritium), Carbon-14, Phosphorus-32, Phosphorus-33, Sulfur-35, Calcium-45, Chromium-51, Cobalt-57, Cobalt-60, Iodine-125, and Barium-133.

In the 15 buildings at the Bethesda and Rockville sites, 603 rooms and areas (defined in this report as survey units) were evaluated to determine the level of required radiological decommissioning efforts. Of the total of 603 survey units, 438 (about 73 %) have been classified as non-impacted and will require no decommissioning efforts. The remaining 165 survey units (about 27 %) have been classified as impacted areas and will require examination for potential radioactive contaminants. The examination for potential radioactive contaminants will include surveys, direct measurements, sampling and analysis, and scanning with appropriate instruments and equipment. Of these 165 impacted areas, one survey unit was classified as Impacted Class 2 and the remaining survey units have been classified as Impacted Class 3. There were no survey sites classified as Impacted Class 1, which is the classification requiring the most extensive decommissioning effort. A randomly selected percentage (numbering approximately 20 to 40 additional survey units) of NMRC common areas, restrooms, and non-impacted areas will be evaluated for potential radioactive contamination. If contamination is found, the classification may be revised. Every reasonable effort will be taken to reduce the contamination to background levels, which would be well below the regulatory release criteria.

It is anticipated that if radioactive contamination is found in NMRC survey units, that contamination would be confined to small areas within a small percentage of the survey units. Sink traps and floor drains will be likely areas for detection of elevated activity. Significant resources are required to accomplish the actions necessary to satisfy the NRC's release criteria for unrestricted use. The estimated cost of performing the necessary decommissioning efforts is $200,000 to $628,000, depending on the availability of no-cost resources (manpower, instruments, materials, services and equipment). The 6- to 12-month decommissioning period will begin in October 1999 and could be concluded as early as April 2000. Decommissioning efforts at the NMRC Rockville Annex site must be completed before the lease expires in November 1999. NMRC Bethesda buildings when vacated in March or April 2000, will revert back to the ownership of NNMC. Future occupants of NMRC Bethesda buildings may include NNMC, the National Institutes of Health (NIH), the Stanley Foundation, and the Uniformed Services University of the Health Sciences (USUHS).

Additional and supportive information is provided as an extensive collection of an Organizational Charts, Floor Diagrams, Documents, Figures, and Tables. Lists for this collection of information are provided following the reference section of the Historical Site Assessment.

E-2. PURPOSE OF THE HISTORICAL SITE ASSESSMENT

In some cases, where sealed sources or small amounts of radionuclides are described by the HSA, the site may qualify for a simplified decommissioning procedure. The HSA:

- Identifies potential, likely, or known sources of radioactive material and radioactive contamination based on existing or derived information
- Identifies sites that need further action as opposed to those posing no threat to human health

- Provides an assessment for the likelihood of contamination migration

- Provides information useful to scoping and characterization surveys

- Provides initial classification of the site or survey unit as impacted or non-impacted

The NMRC Radiation Safety Officer (RSO) has prepared this Historical Site Assessment (HSA) in support of the closure and radiological release of the NMRC sites at Bethesda, Maryland and Annex at Washington Avenue, Rockville, Maryland. This HSA focuses on those NMRC facilities slated for transfer, closure, decontamination, and decommissioning as pertaining to the legislated Base Realignment and Closure (BRAC) requirements. These BRAC activities involve closing down operations at Bethesda and moving most of those operations to the new facility at Forest Glen, Maryland. NMRC will be a co-tenant with Walter Reed Army Institute of Research (WRAIR). Walter Reed Army Medical Center (WRAMC) will become the new Host Command. Since 1942, the NMRC has been actively involved in various types of medical research of interest to DOD. Medical studies have included radiation studies, diving, biological defense, immune cell biology, infectious diseases, resuscitative medicine, environmental stress, aviation medicine, and nutrition research. Medical studies and biomedical research have been conducted which involved the use of chemicals and other hazardous materials, biological materials, animals and animal products, and radiological materials. Since 1987, the NMRC has leased the privately held industrial property from a commercial realty management group. The (NMRC Rockville) Annex is located at 12300 Washington Avenue, Rockville, MD and is housed in a 19,000-square foot, two-story masonry building. The first floor contains administrative offices and research laboratories. The second story is nearly entirely comprised of research laboratories. The Annex houses laboratories involved in enteric and malarial research. Medical studies and biomedical research have been conducted which involved the use of chemicals and other hazardous materials, biological materials, animals and animal products, and radiological materials. All the surrounding properties are located within Rockville city limits and are predominantly for industrial and commercial uses. (reference 10.3, EPA 1997)

Another local, off-site location at which NMRC uses radioactive materials is the NMRC Bone Marrow Registry Program located at 5516 Nicholson Lane, Kensington, Maryland. The NMRC Bone Marrow Registry Program occupies several rooms and laboratories in the leased building at 5516 Nicholson. Limited quantities of radioactive materials had been used in the facility before NMRC began using this building. Initial radiation surveys conducted by the NMRC Radiation Safety Office confirmed that no residual radioactivity was present in the one laboratory in this building in which NMRC started using radioactive materials in 1998. The NMRC Bone Marrow Registry Program will not vacate this building in 1999. Therefore, this building will not require decommissioning efforts by NMRC during 1999.

E-3. PROPERTY IDENTIFICATION -- CHARACTERISTICS

E-3.1 NAME. Naval Medical Research Center (NMRC); the Naval Medical Research Center (NMRC) was known as the Naval Medical Research Institute (NMRI) during October 1942 to September 30, 1998. Located as a tenant on the National Naval Medical Center (NNMC) complex in Bethesda, Maryland. **Appendix B** provides the FY-99 NMRC Organizational chart. NMRC's name and unit identification change took effect on October 1, 1998. **Table E-1** provides a brief description of NMRC buildings and their historical uses.

E-3.2 MISSION. NMRC's mission, as assigned by the Secretary of the Navy, and the tasks to be performed to accomplish this mission, as assigned by the Commander, Department of the Navy, Bureau of Medicine and Surgery, are to: Conduct research, development, tests, and evaluations to enhance the health, safety, and readiness of Navy and marine Corps personnel in the effective performance of peacetime and contingency missions and to perform such other functions or tasks as may be directed by higher authority.

E-3.3 FUNCTION. The functions of the Naval Medical Research Center are to:

- Provide basic and applied research competence in infectious diseases, tissue transplantation, diving and hyperbaric medicine, casualty care, environmental medicine, and human factors directly related to military requirements and operational needs.

- Maintain a program of basic biomedical research in areas of military importance to develop knowledge in anticipation of future problems.

- Provide a scientific potential of the application of new biomedical knowledge to operational problems and requirements.

- Provide a source of scientific advisors and consultants readily available to the operational commands.

- Provide biomedical research capabilities to support field laboratories, naval hospitals, and other naval activities in problems beyond their capabilities.

E-3.4 EXTERNAL COMMAND RELATIONSHIPS.

Current Host: National Naval Medical Center (NNMC)
Support: Bureau of Medicine and Surgery (BUMED)
Area Coordinator: Naval District Washington (NDW)

Future Host: Walter Reed Army Medical Center (WRAMC)
Support: Bureau of Medicine and Surgery (BUMED)
Area Coordinator: Naval District Washington (NDW)

E-3.5 LOCATION. Address: 8901 Wisconsin Avenue, Bethesda, Montgomery County, Maryland 20889-5607. Geographic coordinates: Latitude (North) 39.003078 [39° 0′ 11.1″]; Longitude (South) 77.089600 [77° 5′ 22.6″]; Universal Transverse Mercator (UTM) - Zone 18; UTM X (meters) - 319052.7; UTM Y (meters) - 4318987.5. (reference 10.3, EPA 1997).

E-3.6 FACILITY TYPE. **NMRC Bethesda**. Since 1942, the NMRC has been actively involved in various types of medical research of interest to DOD. Medical studies have included radiation studies, diving, biological defense, immune cell biology, infectious diseases, resuscitative medicine, environmental stress, aviation medicine, and nutrition research. Medical studies and biomedical research have been conducted which involved the use of chemicals and other hazardous materials, biological materials, animals and animal products, and radiological materials. A number of structures that had once been located on the NMRC site have been removed or replaced. Additionally, a number of buildings still in use today have undergone significant changes in their use and many have undergone extensive renovation. **NMRC Rockville Annex**. Since 1987, the NMRC has leased the privately held industrial property from a commercial realty management group. The (NMRC Rockville) Annex is located at 12300 Washington Avenue, Rockville, MD and is housed in a 19,000-square foot, two-story masonry building. The first floor contains administrative offices and research laboratories. The second story is nearly entirely comprised of research laboratories. The Annex houses laboratories involved in enteric and malarial research. The current lease will expire in November 1999. Medical studies and biomedical research have been conducted which involved the use of chemicals and other hazardous materials, biological materials, animals and animal products, and radiological materials. All the surrounding properties are located within Rockville city limits and are predominantly for industrial and commercial uses. (reference 10.3)

E-3.7 TOPOGRAPHY. The NMRC site is located in the City of Bethesda, which is just northwest of Washington, D.C. This portion of Maryland is in the eastern Piedmont physiographic province, which contains hard, crystalline igneous and metamorphic rock. Bedrock in the eastern part of the Piedmont consists of schists, gneiss, gabbro, and other highly metamorphosed sedimentary and igneous rocks of probably volcanic origin. Elevation at the site is approximately 275 feet above mean sea level. (reference 10.3)

E-4. HISTORICAL APPROACH.

E-4.1 APPROACH AND RATIONALE. After a thorough review of historical-site monitoring data, amounts of radioactive materials (RAM) used, periods of RAM use, and the locations of RAM use, the types of radiological surveys to be performed in conjunction with this D&D effort will be site/location specific. This document should not be used as a complete historical accounting of the research efforts and activities conducted at NMRC. This document focuses on those aspects closely or directly related to NMRC's acquisition, use, storage and disposal of licensed (radioactive) materials during the period 1942 through 1999. The goal of the Naval Medical Research Center is to free release all areas from radiological controls to background levels or to levels well below regulatory release criteria. A procedure for surveying, identifying and removing all radioactive contaminants above background will be developed that will incorporate overlapping and/or redundant surveys of all areas that contained radioactive materials. The NMRC Radiation Safety Officer will coordinate all radiological closeout activities. Cost estimates have been prepared to approximate the level of funding needed to cover expenses such as instrumentation, equipment, bottled industrial gases, calibration services, supplemental manpower, and third-party monitoring surveys and reports. **Table E-12** provides cost estimate information for decommissioning activities, surveys, equipment and instruments. **Table E-13** provides a tentative move schedule. This move schedule relates when NMRC Programs and Departments are scheduled to vacate NMRC buildings and relocate to the new facility at Forest Glen, Maryland. Based on this tentative move schedule, a tentative decommissioning schedule has been developed. Schedule changes are anticipated due to the unpredictable nature of the relocation of laboratories, offices, and equipment to a new off-site facility.

E-4.2 BOUNDARIES OF SITE. The NMRC is located in Bethesda, Maryland and operates as a tenant command of the National Naval Medical Center (NNMC). Bethesda lies west of Washington, D.C. and is in Montgomery County. The entire NNMC complex comprises approximately 243 acres and is bounded by Wisconsin Avenue to the west, Jones Bridge Road to the south, a school and residential area along the northern border, Interstate 495 to the northeast, and a sparsely developed, wooded area to the east. Of the 243 acres, the NMRC itself totals no more than 10 acres. The NMRC site is primarily surrounded by other properties owned by the U.S. Government. These include the NNMC and the National Institutes of Health (NIH). Immediately to the southeast of the NMRC complex is a small stream, Stoney Creek, which ultimately drains into the Potomac River. The approximate latitude and longitude are 39.0031 and 77.0896 degrees, respectively. Some of NMRC's research efforts are conducted in an off-site location. NMRC Rockville Annex research efforts are conducted in a leased privately held, two-story building at 12300 Washington Avenue, Rockville, Maryland. The enteric and malarial research programs at the Annex will relocate to the new facility at Forest Glen, Maryland. NMRC will perform D&D efforts at this facility before the lease expires in November-December 1999. The boundary of the survey sites will be limited to the buildings and not to the surroundings grounds. The one exception to this boundary plan is the inclusion of the immediate grounds surrounding the NMRC Building 150, the former Cobalt-60 Irradiation Facility.

E-5. HISTORY AND CURRENT USAGE

E-5.1 HISTORY. The Naval Medical Research Center was commissioned on 27 October 1942 and assigned a general mission of basic, applied and developmental research relating to the health, safety and efficiency of naval personnel. Its original staff, mostly military, was concerned with immediate and practical problems of World War II operations. Extensive studies were conducted on protective clothing, insect repellants, desalination of sea water, aviation oxygen equipment, physiological effects of tropical environments and the effects of immersion in cold water. Other investigations concerned vaccines, night vision, body armor, nutrition, oral hygiene and tropical diseases. After the war, a number of civilian scientists were added to the staff and the research program became more a blend of basic and applied research. Following the pattern envisioned by those mostly responsible for its genesis, the Center headed towards being a large multi-discipline laboratory with military and civilian scientists working on equal footing to provide the Naval Medical Department with a scientific center of international reputation. Rapid developments in nuclear weapons and power dictated the need for a build up of competence in nuclear physics and radiobiology with the major naval capability for medical support centered at NMRC. Before and during the 1960's, NMRC originated proposals to obtain further radiation sources and to extend studies on the biological effects of protons, neutrons, and very short half-life isotopes. These efforts culminated in the decision to create the Tri-Service Armed Forces Radiobiology Research Institute (AFRRI) on the NNMC Bethesda campus in 1961. This creation and other factors led to the demise of all major programs on radiation at NMRC. Radioactive material use at NMRC since the early 1960's have involved the used of sealed-source irradiators and check sources and a variety of unsealed licensed materials. **Table E-1** provides a summary of the historical and current use of NMRC buildings.

E-5.2 HISTORIC OWNERSHIP. Because the NMRC is a tenant on NNMC property, there is no separate legal description of the site found in local government documents. A review of the parcels that comprise the NNMC indicates, however, that NMRC is located primarily on parcel 639, with a small wooded section adjacent to Stoney Creek located on parcel 772. All of the surrounding properties are located within Bethesda city limits. At the time of the NMRC site development, the area was wooded and bordered on the north by grassy fields or small farm plots as indicated in the 1937 aerial photograph provided as AP-1 in the Aerial Photographs section of this HSA. (reference 10.3)

Table E-4 provides information about the use of NMRC buildings and rooms. **Floor Diagrams FD-01 through FD-38** provide floor diagrams for the existing NMRC buildings.

E-5.3 CURRENT USAGE AND OPERATING HISTORY

E-5.3.1 LICENSING AND OPERATIONS.

Prior to 1986, the NNMC Radiation Safety Officer oversaw NMRC's uses of radioactive materials. NMRC records indicate that an Atomic Energy Commission (AEC) radioactive materials license was issued to NMRC for its cobalt-60 facility. This AEC license was terminated in 1963. NMRC maintained NRC licenses for its three cesium-137 gamma irradiators during the 1980's and 1990's. Since 1987, NMRC has used radioactive materials under the conditions of its own Navy Radioactive Materials Permit (NRMP), as part of the Navy's NRC Master Material License. The conditions of NMRC's current NRMP will be amended in March or April 2000 to remove two authorized locations from NMRC's NRMP. These locations are NMRC Bethesda and the NMRC Rockville Annex site.

NMRC has used licensed materials since its operations began in 1942. NMRC possessed an Atomic Energy Commission (AEC) materials license for its Cobalt-60 Facility at NMRC building 150. This AEC By-product Materials License, Number 19-2891-3, was issued on October 2, 1957 and expired on 31 March 1963. Until 1987, NMRC's use of radioactive materials was controlled by NNMC's NRC License, Number 19-02891-05). NMRC held its own NRC License, Number 19-02891-06 for the three cesium-137 gamma irradiators it possessed. NMRC's 9000-Curie irradiator was obtained in 1981; the 2500-Curie irradiator in 1986; and the 300-Curie irradiator in 1992. Beginning in 1987, the Navy adopted its Navy Radioactive Material Permit (NRMP) Program. NMRC's use of unsealed and sealed sources were covered in NRMP Number 19-64223-41NP (initial permit application dated 29 January 1988). When NMRC's name changed from the Naval Medical Research Institute on October 1, 1998, NRMC's NRMP number was changed to 19-32398-41NP to reflect the change in NMRC's Navy unit identification code. NMRC will terminate use of licensed material at the NMRC Bethesda site (March-April 2000) and the Rockville Annex site (November-December 1999).

E-5.3.2 LICENSED (RADIOACTIVE) MATERIAL USE. Historically, NMRC has used a variety of licensed materials, ranging from (1) sealed sources in irradiators and check sources, (2) unsealed radionuclides in liquid form, (3) radioactive gases, (4) radioactive foils, and (5) radioactive microspheres. Since 1992, NMRC's use of radioactive materials have been limited to use of Cesium-137, sealed-source gamma irradiators and unsealed, liquid-form and gas-form radionuclides (H-3, C-14, P-32, P-33, S-35, Ca-45, Cr-51, and I-125).

The radionuclides of contaminant concern are Hydrogen-3, Carbon-14, Phosphorus-32, Phosphorus-33, Sulfur-35, Calcium-45, Chromium-51, Cobalt-57, Cobalt-60, Iodine-125, and Barium-133. **Tables E-3 through E-7** in Appendix E provide data on the historical use of radionuclides at NMRC. **Table E-3** lists the radioisotopes used at NMRC during 1942 through 1999. **Table E-4** (sorted by year of use and by isotope) provides a historical review of the use of the radioisotopes. Using the data provided in **Tables E-3 and E-4**, certain radionuclides that have been used at NMRC will not be included in the list of radionuclides of concern because more than 10 half lives have elapsed since their last use at NMRC.

The isotopes that are not of contamination concern at NMRC (with year of last use, half life (HL), and approximate number of elapsed half lives) are:

Argon, Ar-41 (1992; HL = 1.83 hours; elapsed HL > 30,000);
Cerium, Ce-141 (1993; HL = 32.5 days; elapsed HL > 60);
Cesium, Cs-137 (sealed sources only and negative leak test results) (1999; HL = 30 years);
Iron, Fe-59 (1992; HL = 44.6 days; elapsed HL > 50);
Iodine, I-123 (1991; HL = 13 hours; elapsed HL > 5,300);
Iodine, I-131 (1991; HL = 8.1 days; elapsed HL > 360);
Indium, In-111 (1990; HL = 2.8 days; elapsed HL > 1,170);
Krypton, Kr-79 (1992; HL = 34.9 days; elapsed HL > 70);
Krypton, Kr-83m (1992; HL = 1.9 hours; elapsed HL > 32,200);
Krypton, Kr-85m (1992; HL = 4.4 hours; elapsed HL > 13,900);
Niobium, Nb-95 (1993; HL = 35 days; elapsed HL > 60);
Ruthenium, Ru-103 (1993; HL = 39.5 days; elapsed HL > 50);
Scandium, Sc-46 (1992; HL = 83.8 days; elapsed HL > 30);
Selenium, Se-75 (1991; HL = 119.8 days; elapsed HL > 20);
Tin, Sn-113 (1992; HL = 115 days; elapsed HL > 20);
Strontium, Sr-85 (1992; HL = 64 days; elapsed HL > 39);
Technetium, Tc-99m (1992; HL = 6.05 hours; elapsed HL > 10,000);
Xenon, Xe-127 (1992; HL = 36.4 days; elapsed HL > 70);
Xenon, Xe-133 (1992; HL = 5.3 days; elapsed HL > 480); and,
Ytterbium, Yb-169 (1986; HL = 31.97 days; elapsed HL > 148).

For a survey unit, any detected radioactivity of nuclides of concern will be considered as residual contamination and not as background radiation; exceptions will be noted. All survey units with residual contamination measurements above the NMRC radionuclide-specific investigational level (a small fraction of the regulatory limit) will be subjected to remedial actions and additional surveys. Radioactive material use in NMRC facilities will cease in December 1999. All leak test results (1981-1999) for the sealed-source, cesium-137 irradiators revealed no residual contamination. Except for possible residual cobalt-60 contamination in and around building 150, it is anticipated that no significant quantity of residual radioactive contamination above background levels will be found during the final status surveys.

A review of NMRC's historical and current radiation safety documents for radioactive materials use has been completed for the period 1957 through 1999. Records of use of radioactive materials before this period were not readily available. A review of the records for the period 1957 through 1999 confirm that radioactive materials were used and/or stored in several NMRC (formerly NMRI) laboratories, areas, and buildings. Summaries of the radioisotopes used at NMRC are provided in the table below. **Table E-3** summarizes NMRC radionuclide information (radioisotopes, half life, and major radiation emissions and energies). An in-depth historical review of use of radioactive materials at NMRC during the period 1954 through 1999 was completed. **Table E-4** provides information such as the beginning and ending years of use, isotopes used, maximum on-hand quantities, building and room numbers, investigators and some comments. Personnel were monitored for their occupational exposure to ionizing radiation at NMRC and personal radiation exposures were maintained as low as reasonably achievable. Summaries of annual posted dosimeter results showed that exposure to the general public (non-radiation workers) were not subjected to ambient exposure rates above the 2-mrem/hour limit nor was it likely that the annual dose to any member of the general public could exceed 100 mrem for the year.

Table E-3: RADIOISOTOPES USED AT NMRC (1942 - 1999)

RADIOISOTOPE		HALF LIFE	MAJOR RADIATION	RADIATION ENERGY (keV)
SYMBOL	NAME			
H3	HYDROGEN-3	12.3 Y	BETA	18.6 (max E)
C14	CARBON-14	5730 Y	BETA	156 (max E)
P32	PHOSPHORUS-32	14.3 D	BETA	1710 (max E)
P33	PHOSPHORUS-33	24.4 D	BETA	248 (max E)
S35	SULFUR-35	87.4 D	BETA	167 (max E)
Ar41	ARGON-41	1.83 H	BETA	E-98 (99.2%) 2492 (0.8%)
			GAMMA	1293 (99.1%)
Ca45	CALCIUM-45	165 D	BETA	256 (max E)
Sc46	SCANDIUM-46	83.8 D	ETA	357 (max E)
			GAMMA	889 E-20
Cr51	CHROMIUM-51	27.7 D	GAMMA	320 (10%)
Co57	COBALT-57	270.9 D	GAMMA	122 (85.5%) 136 (E-%) 692 (0.16%)
Fe59	IRON-59	44.6 D	BETA	273 (45%) 466 (53%)
			GAMMA	1099 (56.5%) 1291 (43%)
Se75	SELENIUM-75	E-9.78 D	GAMMA	121 (16.7%) 136 (59%) 264 (60%) 280 (25%) 400 (E-%)
Kr79	KRYPTON-79	34.9 D	GAMMA	136 (0.7%) 261 (9%) 398 (10%) 5E- (15%) 606 (10%)
Kr85m	KRYPTON-85m	4.4 H	BETA	840 (78.6%)
			GAMMA	305 (14%) 151 (75%)
Sr85	STRONTIUM-85	64 D	GAMMA	514 (99%)

Table E-3: RADIOISOTOPES USED AT NMRC (1942-1999) (continued)

RADIOISOTOPE		HALF LIFE	MAJOR RADIATION	RADIATION ENERGY (keV)
SYMBOL	NAME			
Nb95	NIOBIUM-95	35 D	BETA	160 (max E)
			GAMMA	765 (100%)
Tc99m	TECHNETIUM-99m	6.05 H	GAMMA	140 (89%)
Ru103	RUTHENIUM-103	39.5 D	BETA	226 (90%)
			GAMMA	497 (88.9%) 610 (5.6%)
InE-1	INDIUM-E-1	2.8 D	GAMMA	171 (90%) 245 (94%)
SnE-3	TIN-E-3	E-5 D	GAMMA	255 (2.1%) 24 (61%)
I125	IODINE-125	60.2 D	GAMMA	27 (E-0%) 31 (25%)
Xe127	XENON-127	36.4 D	GAMMA	172 (22%) 203 (65%) 375 (20%)
I131	IODINE-131	8.1 D	BETA	606 (90%)
			GAMMA	637 (6.5%) 364 (82%)
Ba133	BARIUM-133	7.2 Y	GAMMA	302 (14%) 356 (69%) 382 (8%)
Xe133	XENON-133	5.3 D	BETA	346 (99%)
			GAMMA	31 (39%) 81 (36%)
Cs137	CESIUM-137	30 Y	BETA	5E- (94.6%)
			GAMMA	662 (90% from Ba137m)
Ce141	CERIUM-141	32.5 D	BETA	434 (70%) 580 (30%)
			GAMMA	145 (48%)
Yb169	YTTERBIUM-169	31.97 D	GAMMA	177 (21%) 198 (35%) 308 (E-%)

E-5.3.3 SEALED SOURCES.

E-5.3.3.1 One Cobalt-60 Gamma Irradiator Facility, NMRC building 150, was operated during 1950-1963; operations were discontinued following a contamination event; residual contamination may be present. **Cobalt Facility Information**: 2500 Curies (Ci) of Cobalt-60 (Co-60), produced by the nuclear pile at Oak Ridge National Laboratory (ORNL) from Co-59, was divided into 60 encapsulated portions, 40 of which were of 50 Ci strength, and 20 of which were of 25 Ci strength. Each of these capsules was contained within a pneumatic capsule that was further contained in a pneumatic carrier tube. The tube system, through application of either positive or negative pressure, connected two radiation chambers separated by a radiation shield. This arrangement permitted either of the two chambers to be entered whenever the RAM was pneumatically transferred to the other. The equipment and sources were housed in a reinforced concrete building that was divided by a heavy radiation shield into two radiation exposure rooms and a control room. The building is constructed of reinforced concrete one foot thick. It is covered with an overhead reinforced concrete slab of concrete 10 inches thick. The control room is separated from the two radiation rooms by a massive radiation shield 3 feet 10 inches thick, constructed of reinforced barite concrete which is 45 pounds heavier per cubic foot than ordinary concrete. This aggregate was used to decrease the size, hence the cost, of the building. Two periscopes pierce the radiation shield, one for each radiation room. The radiation shield between rooms 1 and 2 consisted of a loose fill of barite and scrap iron surrounding the center portion of the tubes of the pneumatic tube transfer system. A 3-feet barite concrete shield, in addition to the ordinary reinforced concrete building walls further protected the radiation room 1, in which the cobalt capsules were stored. Loading and changing of capsules was accomplished in room 1 through the use of a robot carried on a monorail, with visual control conducted through the periscope. Transfer of capsules was completed between rooms in about 6 seconds.

Operator safety features included a warning gong and a safety switch to stop all transfers until persons evacuate the radiation room chamber. In addition, panel lights in the control room indicated capsule positions, interlock switches on both doors disrupted operations until closed, and an audible monitor sounded for each room to signal completion of source transfer. **Floor Diagram FD-38** provides a floor diagram of the remaining ground floor of building 150 and the surrounding grounds.

E-5.3.3.2 Two Cesium-137 Gamma Irradiators located at different NMRC sites, with activities of 300 and 9000 Curies, are used by NMRC researchers and technicians to irradiate specimens and animals. No human irradiations were performed. In August 1998, a third unit containing 2500 Curies of cesium-137, was disposed and removed from the command by Shepherd & Associates, Incorporated.

E-5.3.3.3 Leak Test Results. The leak test results for all Cs-137 gamma irradiators were negative for all periods monitored during 1980-1999. No residual contamination is suspected in NMRC buildings or on the grounds. NMRC's sealed sources consisted of cesium-137 gamma irradiators. Other NMRC sealed sources include a few detection-meter check sources, a few microcurie-strength calibration source rods, and two Radium-226 pellets that were removed from excess counting equipment before disposal.

E-5.3.4 WASTE MANAGEMENT PRACTICES AND STORAGE AREAS. Researchers and technicians using radioactive materials were required to maintain separate waste containers for each isotope used. Further segregation occurred for liquids, solids, carcasses, scintillation vials, and sharps. Since all radioactive material (RAM) wastes were turned in to the Radiation Safety Office for tracking, storage and subsequent disposal and/or transfer. Burial of waste was prohibited. Sewer disposal of liquid aqueous liquids was permitted in most laboratory sinks. During the early 1990's, sewer disposal was restricted to disposal in only two designated sinks. Solid waste disposal via decay-in-storage occurred for shorter half-life RAM (< 65 days). Long half-life RAM wastes were transferred to NNMC for transfer to a commercial contractor. Since 1992, dilute, aqueous RAM waste is collected from the laboratory operations by researchers and technicians and turned in to the NMRC Radiation Safety Office. The Radiation Safety Office then disposes of the liquid waste in one of two designated sinks. Solid wastes are packaged by the researchers and technicians and turned in to the Radiation Safety Office for storage and subsequent disposal. NMRC records reviewed included summaries of NMRC's annual receipts (inventories) of radioactive materials), annual solid radioactive waste disposal logbook entries, and annual sanitary sewer disposal of liquid radioactive materials logbook entries.

E-5.3.5 SPILLS AND CONTAMINATION EVENTS

E-5.3.5.1 In 1962 at the NMRC Cobalt-60 Facility (Building 150), an animal and specimen irradiation facility, a contamination event was discovered. The facility had a sealed-source activity loading totaling 2500 curies consisting of twenty 25-Ci clay-encapsulated sources and forty 50-Ci ceramic-encapsulated sources. The discovery of the contamination occurred following an investigation of abnormally personal internal monitoring results; operations were ceased during 1962. The sources were removed and sent to the ORNL. Decontamination events involved removing materials from the site, removing several inches of the soil from the fenced vicinity of the building and packaging and shipping contaminated wastes to an authorized Army disposal site. The termination of the AEC License Number 19-2891-3 on 31 March 1963 occurred before all decontamination efforts had been completed. Residual radioactivity remains at the site; the amount and type are undetermined and requires characterization before the area can be assigned a radiological classification. The 1962 discovery of cobalt-60 contamination led to the halting of all cobalt-60 operations in 1962, removal of all sources in 1963, the decontamination efforts, and the expiration of the NMRC Atomic Energy Commission license in March 1963.

E-5.3.5.2 In 1993, an incident occurred at the Rockville Annex involving the unauthorized use and spill of P-32, inadequate post-use monitoring, and non-reporting of broken glass tubes containing P-32 during the use of a laboratory centrifuge unit. Contamination was spread to various locations at the site; decontamination events occurred to clean up as much as possible and the residual radioactivity was covered until radiation decay occurred; no other spills occurred since then. No residual radioactivity remains at the Rockville site in 1999 from that 1993 spill.

E-5.3.5.3 Leakage of Xenon-127 gas in the hood of room E-2 of building 53 was discovered during October 1985. The leakage was attributed to a leaky connection between the container and the gas-tight syringe used to draw gas doses from the container. It was suspected that another contributor to the leakage was the non-closure of the container valve after a gas sample was withdrawn from the container. The leakage was estimated to be between 28 mCi and 100 mCi over a 9-week period, depending on the actual time of loss. No decontamination efforts were required. No residual contamination remains at the building 53 site as a result of this leakage.

E-5.3.6 USE OF URANYL ACETATE/NITRATE

Uranyl acetate was obtained approximately 10 years ago and has been used in two labs in building 17B (Rooms 15 and S-30) periodically over the last 10 years. U-238 (depleted) was used in the form of uranyl acetate or uranyl nitrate and was not high enough in radioactive content to be included in NMRC's Permit (since 1988) or on NNMC's NRC license (prior to 1998). This material was not considered a licensed material. The material was purchased as a powder or pellets and was mixed with liquids to fixate slides in laboratory procedures. Possible means of disposal included disposal down the sinks in the labs or as bio-hazardous waste. There were no documented spills of this material; at no time were the materials under pressure that could have resulted in a spread of contamination to the walls and ceilings; there was a potential for contamination of this unsealed material on the countertops and at the sinks. A protocol and reference information describing the use of uranyl acetate or uranyl nitrate in two NMRC laboratories in building 17 was reviewed. The current inventory of uranyl acetate is six bottles containing approximately 25 grams each.

E-5.4 ADJACENT LAND USAGE. The NMRC site is primarily surrounded by other properties owned by the U.S. Government. These include the NNMC and the National Institutes of Health (NIH). Immediately to the southeast of the NMRC complex is a small stream, Stoney Creek, which ultimately drains into the Potomac River. The approximate latitude and longitude are 39.0031 and 77.0896 degrees, respectively. **Figure F-01** provides the Site Location map for NMRC. **Figure F-02**, the Site Resource map, presents the layout of buildings, both past and present, at the NMRC site. Also included in this Historical Site Assessment (HSA) is the NMRC Rockville Annex. NMRC Rockville Annex research efforts are conducted in a leased, privately held, two-story building at 12300 Washington Avenue, Rockville, Maryland. This Annex building is on a remote, isolated site in a commercial area.

E-5.5 FUTURE USAGE. Control of the NMRC buildings at NNMC in Bethesda and at the Rockville Annex will revert back to owners or property managers. The lease for the NMRC Rockville Annex building will likely expire during December 1999. NMRC research efforts will move from the building during June and October 1999. Control of the building will revert to its commercial management firm. The upper levels of building 150 were removed in the 1960's. The building has not been used or occupied since that period. During February-April 2000 following decommissioning actions, the remains of building 150 and the surrounding grounds will revert back to the owner, NNMC. During February-April 2000, control of NMRC buildings 17, 18, 21, 29, and 49 (and possibly others) will revert back to NNMC ownership before they are turned over for major renovations then occupancy by components of the National Institutes of Health (NIH). During February-April 2000, control of NMRC building 53 (and possibly others) will revert back to NNMC ownership before they are turned over for major renovations then occupancy by components of the Uniformed Services University of the Health Sciences (USUHS).

E-6. FINDINGS

E-6.1 POTENTIAL CONTAMINANTS

A review of NMRC and NNMC radiation safety records, licenses and permits, radionuclide receipt and disposal logs, and on-hand inventory records has been completed. This review indicated that the potential radioactive contaminants at the NMRC site would be some of those radionuclides listed in **Table E-5.**

Table E-5 provides a list of potential contaminants in NMRC buildings and was compiled using the radionuclide receipt and disposal logs and inventory records. The potential radioactive contaminants include Hydrogen-3 (tritium), Carbon-14, Phosphorus-32, Phosphorus-33, Sulfur-35, Calcium-45, Chromium-51, Cobalt-57, Cobalt-60, Iodine-125, and Barium-133. Other tables provided in this report specify the contaminants for particular NMRC rooms and spaces.

Table E-5. Potential Radioactive Contaminants and Radiological Classifications (1942-1999)

BUILDING NUMBER	BUILDING DESCRIPTION OR USE	RADIOLOGICAL CLASSIFICATIONS	FLOOR AREA (SQUARE FOOTAGE)	OCCUPANCY OR YEARS OF RAM USE		ISOTOPES OF CONCERN
				Start	Stop	
17 17A 17B	Offices, Labs, Storage, Utility Machinery	Non-impacted and Impacted Class 3	84,398 (total)	1942 1946 1956	1999 1999 1999	H3,C14,P32, P33, S35, Ca45, Cr51, Co57, I125, Ba133
18	Animal Facility, Model shops, Offices, Labs, Storage	Non-impacted and Impacted Class 3	14,300	1942	1999	H3, C14, P32, S35, Cr51
21	Animal Facility, Offices, Labs, Storage	Non-impacted And Impacted Class 3	35,828	1946	1999	H3,C14, P32, P33, S35, Ca45, Cr51, I125
28	Machine Shop, Offices, Labs, Storage	Non-impacted And Impacted Class 3	5,856	1952	1999	H3,C14 (highly unlikely)
29	Offices, Labs, Storage, X-ray Facility	Non-impacted And Impacted Class 3	630	1955	1999	H3,C14 (highly unlikely)
53	Diving Facility, Offices, Labs, Storage	Non-impacted And Impacted Class 3	35,000	1976	1999	H3,C14,P32, P33,S35, Ca45, Cr51, Co57, I125, Ba133
59	Diving Facility	Non-impacted	2,695	1989	1999	None
69	H2/O2 Research Facility	Non-impacted	805	1992	1999	None
79	Equipment Facility	Non-impacted	1,195	1992	1999	None
139	Offices, Labs, Storage	Non-impacted And Impacted Class 3	4,343	1944	1999	H3,C14, I125 (very unlikely)
150	Cobalt-60 Irradiator	Impacted Class 2 and 3 Areas	2,380	1950	1963	Co-60
ANNEX	Offices, Labs, Storage	Non-impacted And Impacted Class 3	10,657	1986	1999	H3,C14,P32, S35,Cr51
ALL OTHERS	Offices, Labs, Storage	Non-impacted			--	(highly unlikely)

E-6.2 POTENTIAL CONTAMINATED AREAS

E-6.2.1 PREVIOUS RADIOLOGICAL SURVEY RESULTS. All available NMRC daily, weekly, monthly, and quarterly wipe test and survey monitoring results (1979-1999) were reviewed for evidence of residual contamination in NMRC sites. Incident and spills records were also reviewed. A significant spill involving phosphorus-32 (P-32) occurred at the NMRC Rockville Annex in October 1993. No residual radioactivity remains because of the 1993 spill. More than 100 half-life for P-32 have elapsed since that incident. There have been no other incidents of radioactive material spills at the Rockville Annex site. There have been no significant spills or incidents involving radioactive materials at NMRC Bethesda since the 1962 event involving cobalt-60 at and around NMRC building 150. NMRC decommissioning efforts will include final surveys for the Building 150 site and surrounding grounds. With the possible exception of the building 150 site, previous survey data for NMRC sites indicate that there is no reasonable expectation to find residual radioactivity that greatly exceeds background levels. It is anticipated that no NMRC survey unit will have evidence of residual radioactivity that exceed unrestricted-release-criteria guideline values. The NMRC survey sites selected for scoping surveys are all sinks and traps and the building and grounds of Building 150. If contamination is found, characterization surveys, remedial actions, and remedial action surveys will be completed before the final surveys for these sites are initiated.

Table E-5 provides a list of potential contaminants and radiological classifications for NMRC buildings. This list was compiled using the radionuclide receipt and disposal logs and inventory records. **Table E-11** provides a list of the NMRC locations for sewer disposal of liquid radionuclide wastes.

Table E-6 provides a summary of the sites and periods of radiological wipe testing and survey meter monitoring. The table indicates those NMRC buildings and rooms that were used for RAM work and were subjected to periodic radiation monitoring. The purpose of the monitoring was real-time detection of residual contamination in the laboratories and on individuals. Once residual contamination was detected, immediate actions were taken to reduce the residual activity to background levels. The NMRC Radiation Safety philosophy has been to maintain a residual radioactivity ALARA and/or at background levels. Records reviewed to demonstrate the level and extent of residual contamination included actual NMRC quarterly radiological survey monitoring and wipe test results (by building and room numbers), summaries, and laboratory layout diagrams. When areas of elevated activity (contamination as determine by readings more than twice background levels) were detected, the Radiation Safety Office took immediate actions to decontaminate and re-monitor these areas until background radiation levels were achieved.

A review of calibration certificates for NMRC Radiation Monitoring and Detection Equipment and Instruments for the period 1995-1999 confirmed that radiation detection and monitoring equipment were properly maintained and calibrated.

E-6.2.2 IMPACTED AREAS - KNOWN AND POTENTIAL. Impacted areas have some potential for residual contamination. Impacted areas are further divided into three classifications. **Table E-7** provides the radiological classifications of NMRC building and rooms.

E-6.2.2.1 CLASS 1 AREAS. Class 1 areas are impacted areas that, prior to remediation, are expected to have concentrations of residual radioactivity that exceed the $DCGL_W$ ($DCGL_W$ is defined in MARSSIM). NMRC characterized none of its survey sites as Class 1.

E-6.2.2.2 CLASS 2 AREAS. Class 2 areas are impacted areas that, prior to remediation, are not likely to have concentrations of residual activity the exceed the $DCGL_W$.

E-6.2.2.3 CLASS 3 AREAS. Class 3 areas are impacted that have a low probability of containing residual radioactivity.

E-6.2.3 NON-IMPACTED AREAS. Non-impacted areas have no reasonable potential for residual contamination.

E-6.2.4 BUILDING DESCRIPTIONS AND CLASSIFICATIONS. The NMRC Radiation Safety Officer completed a historical review of documents of NMRC's use of radioactive materials for the period 1942 through 1999. Interviews were also completed. These reviews and interviews revealed that NMRC has used licensed materials in the majority of the NMRC buildings on the NNMC Bethesda campus and in most of the rooms in a two-story lease building located in Rockville, Maryland. The possession limits for unsealed radioactive materials varied from a few microcuries to several hundred curies for certain short-lived isotopes. The isotopes of contamination concern for the majority of the NMRC survey units will be tritium (H-3), Carbon-14, and Sulfur-35. The isotope of concern for the Building 150 survey unit will be Cobalt-60. No residual Cesium-137 contamination as a result of the gamma irradiator usage is expected because of the negative (semi-annual) leak-test results for the period 1981 through 1999. Based on the history of the radioisotopes used and the periodic room and area surveys conducted by NMRC Radiation Safety, it is not anticipated that any significant quantity above background will be found during the final survey of all NMRC survey sites.

Table E-7 provides the radiological classification for NMRC buildings in which radioactive materials have been used. Demolished buildings and sites are not included in the table. Also listed are 1999 radioactive contaminants of concern. The 1999 contaminants of concern are defined as those radioactive materials, while accounting for radioactive decay, may be present as residual radioactive contamination. **Table E-8** also provides a complete list of NMRC building and room numbers, square footage of floor areas, contaminants of concern, and radiological classifications for those buildings listed in **Table E-7**.

Table E-9 summarizes the NMRC Impacted Class 2 and Class 3 Areas.

Table E-10 summarizes the Non-impacted Areas.

NMRC survey sites are classified as:

- **Non-impacted** (offices and non-laboratory areas),
- **Impacted Class 2** (one site with known residual contamination; building 150; cobalt-60; 1962 contamination event), or
- **Impacted Class 3** (current NMRC licensed-material-use and -storage areas).

E-6.3 POTENTIAL CONTAMINATED MEDIA. The potential contaminated media will consist of building surfaces and soil.

E-6.4 RELATED ENVIRONMENTAL CONCERNS: None

E-7. NMRC CONCEPTUAL MODEL AND SITE DIAGRAM INFORMATION

For the convenience of scheduling necessary radiological activities and to maximize the use of available resources, NMRC's Decontamination and Decommissioning (D&D) Plan has been divided into four distinct phases. These phases may be completed serially or concurrently.

Figures F-01 and F-02 provide diagrams of the NMRC site on the NNMC Bethesda campus. Because only one isolated building is involved, no diagram is provided depicting the NMRC Rockville Annex site. **Floor Diagrams FD-01 through FD-38** provide floor plans for the NMRC Bethesda and NMRC Rockville Annex sites.

Table E-14 provides a summary of the four phases depicted in the Conceptual Model Diagram, **Figure F-03**.

Table E-15 provides regulatory-agency-screening values for direct comparison of survey data to determine if the residual contamination levels meet the release criteria.

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E-6.4 RELATED ENVIRONMENTAL CONCERNS: None

E-7. NMRC CONCEPTUAL MODEL AND SITE DIAGRAM INFORMATION

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Table E-14 provides a summary of the four phases depicted in the Conceptual Model Diagram, **Figure F-03**.

Table E-15 provides regulatory-agency-screening values for direct comparison of survey data to determine if the residual contamination levels meet the release criteria.

Table E-14. NMRC D&D Conceptual Model Information

NMRC D&D Phase	**Buildings**	**Number of Survey Units**		**Floor Area (square feet)**	
		Non-impacted	**Class 2 or 3**	**Non-impacted**	**Class 2 or 3**
1	17, 18, 21, 22, 29, 49, 79, 139, 146, 174, 219	334	117	63,509	32,217
2	Rockville Annex	31	35	4,617	7,595
3	150	0	2	0	2,380
4	28, 53, 59, 69	73	11	21,812	6,902
Totals		438	165	89,938	49,094

Figure F-03. NMRC CONCEPTUAL MODEL (June 1999) Phase I of IV

NMRC D&D Phase I (10/1999-3/2000)
Main NMRC Bethesda building numbers:
17,18,21,22,29,49,139, 146,174,219

Contaminants of concern:
H-3, C-14, P-32, P-33, S-35, Ca-45, Cr-51, Co-57, I-125, Ba-133

BUILDING SURFACE CLASSIFICATIONS

	Number Of Survey Units	Total Area (SF)
Non-impacted	334	63,509
Impacted Class 3	117	32,217
Totals	451	95,726

Figure F-03. NMRC CONCEPTUAL MODEL (June 1999) Phase II of IV

NMRC D&D Phase II (10/1999-11/1999)
NMRC Rockville Annex location (1 building)

Contaminants of concern:
H-3, C-14, P-32, S-35, Cr-51

BUILDING SURFACE CLASSIFICATIONS

	Number Of Survey Units	Total Area (SF)
Non-impacted	31	4,617
Impacted Class 3	35	7,595
Totals	66	12,212

Figure F-03. NMRC CONCEPTUAL MODEL (June 1999) Phase III of IV

NMRC D&D Phase III (10/1999-2/2000)
NMRC Building 150 and Grounds

Contaminant: Co-60

SURVEY AREA CLASSIFICATIONS

	Number Of Survey Units	Total Area (SF)
Grounds (Soil):		
Impacted Class 3	1	1,500
Building Surfaces:		
Impacted Class 2	1	880

Figure F-03. NMRC CONCEPTUAL MODEL (June 1999) Phase IV of IV

NMRC D&D Phase IV (10/1999-2/2000)
NMRC Bethesda building numbers: 28,53,59,69

Contaminants of concern:
H-3, C-14, P-32, P-33, Ca-45, Co-57, Ba-133

BUILDING SURFACE CLASSIFICATIONS

	Number Of Survey Units	Total Area (SF)
Non-impacted	73	21,812
Impacted Class 3	11	6,902
Totals	84	28,714

Table E-17 provides a Milestone Chart for D&D Final Status Survey Activities. This table shows that the successful completion of the Historical Site Assessment accomplishes the first step in this milestone chart, "Evaluate Contamination Potential". This chart will be updated periodically during the D&D process to track the progress of required D&D efforts.

Table E-18 summarizes the types of radiological surveys that may be performed and documented to satisfy regulatory decommissioning requirements.

Table E-17 provides NRC-approved screening values. These screening values will be compared directly to NMRC's final status survey data to verify compliance with the NRC's release criteria for unrestricted use. Criteria for interpreting sample survey measurements when no reference area is used are provided as **Table E-18.** Using the screening values will eliminate the need to perform pathway dose calculations to satisfy the NRC limit for unrestricted release of no more than 25 mrem/year evaluated over a period of 100 years.

E-8. CONCLUSIONS

A comprehensive Historical Site Assessment (HSA) to support the closure and relocation of the Naval Medical Research Center (NMRC) (formerly the Naval Medical Research Institute, NMRI) has been completed. This HSA will serve as the springboard from which NMRC's Radiological Decontamination and Decommissioning (D&D) efforts will be launched.

Even though radiation safety documents for the period 1942 through roughly 1970 were not readily available for review, NMRC's annual historical reports were useful in shedding light on the type of projects undertaken, where experimentation and work were performed and by whom, and the types/quantities of radioactive materials used. In addition, data for the past two decades were thoroughly reviewed and have been summarized in this HSA.

As was presented in this HSA, NMRC's history of military medical research during 1942 to 1999 at the Bethesda site and during 1986 to 1999 at the NMRC Rockville Annex site included the use of sealed and unsealed radioactive materials. NMRC's sealed-source use involved the use of cobalt-60 during the 1950's and 1960's and the use of cesium-137 in gamma irradiators in the 1980's and 1990's. Residual cobalt-60 contaminants are suspected in and around the remnants of building 150. Required, semi-annual leak test results for the gamma irradiators provide assurance that there is low probability of residual cesium-137 contamination at the NMRC sites. The use of unsealed radioactive materials in more than 150 laboratories, rooms, and areas dictate that some level of decommissioning is required for each area of radioactive material use to ensure and document compliance with the NRC's release criteria for unrestricted use. The contaminants of concern are Hydrogen-3 (tritium), Carbon-14, Phosphorus-32, Phosphorus-33, Calcium-45, Chromium-51, Cobalt-57, Cobalt-60, Iodine-125, and Barium-133.

NMRC's decommissioning efforts will focus on survey sites in the existing NMRC buildings. In addition, a random selection of areas designated as non-impacted will be subjected to the same level of D&D effort as some impacted areas. After equipment, materials and wastes are removed from radioactive-materials-use areas, it is anticipated that the residual contamination will be confined to small areas within a small percentage of the survey units. Sink traps and floor drains will be likely areas for detection of elevated activity.

E-9. REFERENCES

E-9.1 **NMRC HSA, NMRC Historical Site Assessment, June 15, 1999**

E-9.2 **NUREE-1575, MARSSIM,** Multi-Agency Survey and Site Investigation Manual (December 1997).

E-9.3 **ESA 1998,** Environmental Site Assessment, Naval Medical Research Institute (NMRI), by the Environmental Company, Inc., May 1998.

E-9.4 **NRC Nuclear Licensing Reports**, Volume 12, Number 11, D&D Screening Values, November 1998

E-9.5 **Reg Guide 1.86,** Nuclear Regulatory Commission Regulatory Guide 1.86 (circa 1974)